

An AI-Based Adaptive Learning System for Teaching Pythagoras Theorem to Blind and Deaf Students

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Abstract

Learning mathematics still remains a tough task for those students who have sensory disabilities as there are traditional ways of teaching mathematics which are based on visual imagery, videos, and hearing instructions. These ways of teaching are not useful for the visually disabled (blind) and hearing-disabled (deaf). Current systems of support, including transcription, primitive screen reader systems, and sign-based systems, have failed to meet the need of an adaptable system.

To counteract these problems, the following paper suggests the development of an AI-Based Adaptive Learning System for Blind and Deaf Students. It offers personalized learning experience through analysis of user performance through machine learning methods like logistic regression classification. Based on pre-test performance, students will be classified as slow or advanced learners, and the content will be modified based on this assessment. The proposed approach uses Generative AI for chatbot assistance, Text-to-Speech (TTS) for visually impaired users, and visual instructions with captions for hearing-impaired learners.

The Agentic AI module is implemented to make smart decisions on what content should be provided to each student to support an adaptive and personalized approach to learning. Moreover, the use of Explainable AI (XAI) allows for providing feedback on the performance of learners by pinpointing their weaknesses and recommending measures that can help improve their overall performance. The proposed system proves its feasibility through the application of mathematical theories like the Pythagoras Theorem.

Keywords: *Adaptive Learning, Inclusive Education, Blind and Deaf Students, Personalized Learning, Machine Learning, Generative AI, Agentic AI, Explainable AI, Pythagoras Theorem*

I. Introduction

The entitlement to good education is a well-recognized one, but there are many difficulties encountered by those who have sensory impairment when it comes to mathematical learning. Conventional ways of teaching mathematics involve visual aids like diagrams, graphs, and even writings, which cannot be grasped by blind students. Deaf students, on the other hand, struggle to comprehend oral explanations and communications, without which the concepts cannot be fully grasped.

Despite the availability of several assistive technologies including screen readers, Braille technology, and captioning aids that give some form of accessibility, they are inflexible and cannot offer an individualized learning experience to the learner. The rate at which a learner learns cannot be taken into consideration.

In order to tackle these issues, this project recommends the implementation of an AI-Based Adaptive Learning System for Blind and Deaf Students. Machine learning algorithms like Random Forest classification are used to evaluate the performance of the learners and classify them into slow and advanced learners depending on their pre-test results. Depending on the classification done, the content of learning is customized accordingly. Generative AI is introduced by the incorporation of a chatbot that interacts with the learner and answers any questions raised by them.

The Agentic AI module is included to smartly select the best learning path for each individual learner. Furthermore, the use of Explainable AI (XAI) ensures feedback through highlighting weak points and providing improvement suggestions. The system is demonstrated using mathematical concepts such as the Pythagoras Theorem, aiming to improve accessibility, enhance learning outcomes, and promote inclusive education.

II. Literature Review

Artificial intelligence, machine learning, and tools helping students learn have changed quickly, pushing educators to think about learning in a way that includes everyone. Educators are increasingly studying how to make education more accessible using AI-powered adaptive learning, smart tutors, tailored learning paths, and support systems designed for children who are blind or deaf. While studies often improve how well these tools work, many systems still only focus on single activities, ignoring how things like accessibility, behavior analysis, math tailored to student needs could all fit together into a single model.

A. AI-Based Inclusive Education and Adaptive Learning

Starting with a general overview of AI-based educational systems, B. Sumak and co-authors investigated artificial intelligence systems for developing an inclusive education environment. The research identified a few innovative technologies used to enhance education accessibility and learner engagement. Although various artificial intelligence-based systems were identified, no predictive machine learning models were developed for mathematics education purposes.

Moreover, the system did not consider the development of a disability-specific adaptive curriculum generator (B. Sumak et al., 2024).

Very soon, M. J. Alam and co-workers assessed the advantages and disadvantages of AI-based inclusive learning environments. Intelligent automation and personalization support were introduced to enhance accessibility in the field of education. Nevertheless, the authors failed to integrate predictive analytics and classification-driven adaptive routing into the theoretical framework. Mathematics-focused assessment of learners' performance also required further exploration (M. J. Alam et al., 2024).

Simultaneously, V. Tripathi and co-authors presented a review of AI tools for facilitating education among people with disabilities. Their literature review included several innovative tools that could be applied in different educational sectors. Still, predictive learning models for mathematics and disability-specific content delivery systems were not mentioned (V. Tripathi et al., 2024).

With the advent of intelligent tutoring systems, Garcia and Smith proposed AI-based personalized educational platforms for the use of K–12 students. While the adaptive instruction proved efficient to improve learner engagement and personalize educational content, accessibility-aware learning assistance services including Text-to-Speech, animated teaching tools, and voice-based chatbots required for sensory disability learners were omitted from the system (M. Garcia and J. Smith, 2022).

In a similar vein, Kumar and Sharma presented adaptive learning systems based on AI for educational environments. In the framework, the traditional classifier-based personalized learning techniques were employed. Despite the improvement of learner adaptation process, the system only utilized one classifier without incorporating any dedicated component for learners with disabilities such as blindness or deafness. Moreover, deep learning and behavior clustering methodologies were also absent (A. Kumar and P. Sharma, 2022).

Considering inclusive learning systems in general, Verma conducted a study on the machine learning applications in educational accessibility frameworks. As a result, the study demonstrated that intelligent educational systems lacked in providing support for visually impaired and hearing-impaired learners (P. Verma, 2021).

Earlier research carried out by P. Verma also investigated machine learning in inclusive education in its early developmental stages. Back then, disability-based learning assistance was relatively unexplored, and there were no adaptive assistive learning models for visually impaired and hearing-impaired students (P. Verma, 2018).

From these studies, it is evident that AI-driven adaptive learning models enhance accessibility and student engagement to some degree. However, most of the models are still theoretical, do not incorporate predictive analysis for learners, and cannot integrate adaptive mathematics learning and multi-disability educational assistance.

B. Accessibility Technologies for Blind and Deaf Learners

Starting from visually impaired learners, M. H. Al Omoush et al. designed mathematics learning systems enhanced by technologies for use among blind learners. Educational technologies provided more accessibility in terms of mathematics learning as well as instructional delivery. Nevertheless, despite the advancements made, the proposed model focused exclusively on visually impaired learners and did not consider predictive clustering, deep learning classification, and behavioral analysis based on LSTM neural networks (M. H. Al Omoush et al., 2024).

With regard to tactile educational methods, B. L. Due suggested manipulative learning techniques to teach the Pythagorean theorem to blind learners. Physical learning proved beneficial due to greater learner engagement with the topic. Nevertheless, the process still was hardly automatable and scalable, and no tracking of learner progression, trajectory prediction, and algorithmic educational content generation were involved (B. L. Due, 2024).

Concerning audio learning approaches, A. Sharma created speech-enhanced learning frameworks for visually impaired learners. Audio delivery proved quite helpful when working with blind learners. Nevertheless, in this case, no adaptive feedback, personalization, and differentiated instruction depending on learner progress were implemented (A. Sharma, 2020).

In like manner, R. Mehta designed educational frameworks utilizing screen readers for visually challenged university students. The system was enhanced by interface compatibility and digitalized reading capabilities. However, there was a lack of intelligent learner feedback systems as well as the automatic adjustment of the content difficulty in relation to the observable learner behaviors within the system (R. Mehta, 2019).

In the study about accessibility within mathematics, Escobedo et al. integrated the usage of Braille recognition machine vision systems that assist blind children in mathematics education. Numerical understanding was achieved through machine vision and Braille interaction. Unfortunately, the system only had an ability to comprehend two-dimensional mathematical constructs without any provisions for predictive learning capabilities or deaf learning community (L. Escobedo et al., 2019).

Regarding hearing-impaired learners, S. Alnafouri and G. Issa presented an innovative AI-based framework of personalized learning instruction for deaf students. Learning personalization and individualized educational instruction were supported through intelligent instructional adaptation. However, there were no provisions for structured symbolic mathematics learning and performance level classification within the system (S. Alnafouri and G. Issa, 2024).

One of the pioneering researches on education accessibility was carried out by Ferreira and Cavaco, who came up with an auditory tactile learning system in the form of a game targeted at visually impaired students. Interactivity in educational games enhanced the experience of learning basic arithmetic. However, the lack of adaptive difficulty settings meant that fast learners lacked stimulation, while slow learners lacked support (Ferreira and Cavaco, 2014).

C. Machine Learning, Deep Learning, and Education Prediction

Machine learning and deep learning methods are becoming more common in education prediction and intelligent learner profiling. S. Santhiya and his team used machine learning classification methods to identify at-risk learners based on educational data analysis. Learner prediction accuracy improved remarkably. However, there were no adaptive educational path structures or any personalized remediation procedures for the learners classified as being at-risk (S. Santhiya et al., 2023).

Clustering and decision tree methods were among the techniques used to process general student behavior and performance by Johnson and Brown. Data mining methods proved quite useful for processing learners' behavioral data. However, the system was designed to work with non-sensory-impaired students (E. Johnson and R. Brown, 2020).

Sign language recognition techniques have been further researched through deep learning techniques. W. Chen and his team successfully created a system capable of sign language recognition using deep neural networks and showed very good classification accuracy. Communication support capabilities were greatly enhanced. However, the system could only be used for communication assistance and was not an adaptive educational system or intelligent tutoring system (W. Chen et al., 2021).

Learning pathways through knowledge-graph learning techniques have also been discovered. The authors M. Mackowski and others used learning paths based on the knowledge graphs to assist blind learners of universities. The educational navigation was enhanced via sequencing and modeling of the learning pathway structure. However, the use of numerical learning, behavioral clustering, and learner session-level tracking was not introduced in the system (M. Mackowski et al., 2023).

Later, the authors X. Zhao et al. have studied applications of generative AI for assisting learners with disabilities in the university setting. The performance was significantly improved by reading and writing education generated by artificial intelligence. Nevertheless, the mathematics learning, multistep symbolic reasoning, and adaptive education of sensory-disabled learners were not covered in the framework (X. Zhao et al., 2025).

In terms of the mathematics learning analysis, the researchers S. Metikasari et al. reported common misconceptions of slow learners regarding the Pythagorean theorem. Learning problems were identified as well as common misconceptions. An intelligent or technologically-assisted educational solution has not been suggested, moreover, the sample size was quite limited (S. Metikasari et al., 2019).

Despite advancements in the fields of machine learning and deep learning, most frameworks fail to incorporate features such as supervised classification, unsupervised behavioral clustering, multiple session learner trajectory analysis, and adaptive mathematics learning.

III. Research Gaps and Motivation

A systematic review of existing literature uncovers several critical research gaps that motivate the current study.

A. Absence of a Unifying Educational System Supporting Both Visually and Hearing Impaired Learners

Most prior research has been conducted individually to cater to learners with visual impairment or hearing impairment, but the need to develop a single unified adaptive educational platform that caters to learners from both groups remains unfulfilled. This study develops an adaptive educational platform catering to learners with two different disabilities in one place.

B. Absence of Predictive Behavioral Modeling and Trajectory Forecasting

The existing systems are mostly providing static support to learners and do not conduct any analysis to predict their future behavioral trends. There is a dire need to conduct predictive clustering, analyze user behavior, and use LSTM models for trajectory forecasting.

C. Absence of Explainable AI in Inclusive Education

Most intelligent educational systems operate as black-box systems without any explanations of predictions about students' behavior or adaptive recommendations. There is almost no application of SHAP-based Explainable AI methods within educational systems. The presented solution addresses this problem by introducing explainable analysis of learners' behavior.

D. Insufficient Development of Generative AI for Mathematics Accessibility

Generative AI solutions currently implemented for disabled learners primarily focus on reading and writing activities but not on structured mathematics learning. Creation of adaptive mathematical content based on learners' performance and their specific accessibility needs has been overlooked. Generative AI for adaptive mathematics education will be an essential part of the developed solution.

E. Absence of Multi-Stage AI Educational Framework

The majority of existing solutions concentrate exclusively on solving separate educational problems: either creating some accessibility tools, or building an intelligent tutoring system, clustering learners, or classifying learners' behavior. Very few systems have managed to combine all these approaches within a single framework. The presented solution aims to address this deficiency.

IV. Proposed Methodology

A. Proposed System Architecture

The proposed system architecture was designed to provide an adaptive and accessible learning environment for blind and deaf students learning the Pythagoras Theorem. The system personalizes the learning process based on two important factors: the student's disability type and learning capability. Rather than delivering the same information to all learners, the system will adapt the teaching methods, interaction, and explanation levels to suit individual requirements of each learner.

The process starts when the student enters the website and specifies their disability type, whether blind or deaf. According to the input given, the system activates a specific interface that caters to blind and deaf disabilities. The blind learners will be presented with an auditory system using Text-to-Speech technology and voice commands, while the deaf learners will have an interface filled with videos, animations, images, and other visual cues. This helps maintain an effective learning system for both groups of disabled people. After logging in, the student will take a pre-assessment test on Pythagoras Theorem. In this process, the system will gather several behavioral attributes, which include the student's score, response time, number of attempts, correctness rate, and hint request. These behavioral attributes are then fed to machine learning and deep learning models such as Logistic Regression, Random Forest, Naive Bayes, SVM, KNN, ANN, and DNN to classify the students as either Slow or Advanced Learners. Out of all these classifiers, the one with the highest prediction accuracy is Logistic Regression.

The prediction outcome and disability category are sent to the Agentic AI controller, which is the decision-making component of the system. Using the information about the student's profile, the Agentic AI chooses among four different learning strategies:

- Blind Slow Learner
- Blind Advanced Learner
- Deaf Slow Learner
- Deaf Advanced Learner

The blind students get their explanations through the sounds in a Google Gemini-based chatbot, while the deaf learners get their animations and captioned educational material. Slow learners are given more elaborate explanations and guidance, and fast learners get concise and quick content for instruction.

Once the learning process is over, the learners have a final quiz to test their progress. Quiz results are evaluated using the LSTM neural network to monitor the behavior of learners across several lessons and recognize any pattern in the learning process. SHAP explainable AI is then used to recognize what factors led to the classification of learners. Finally, the report of the performance of learners is generated from the results.

The incorporation of concepts such as Machine Learning, Deep Learning, Agentic Artificial Intelligence, Generative AI, LSTM tracking, and Explainable AI will enable this proposed approach to offer intelligent and adaptive education for blind and deaf students.

Fig 1. System Architecture Diagram

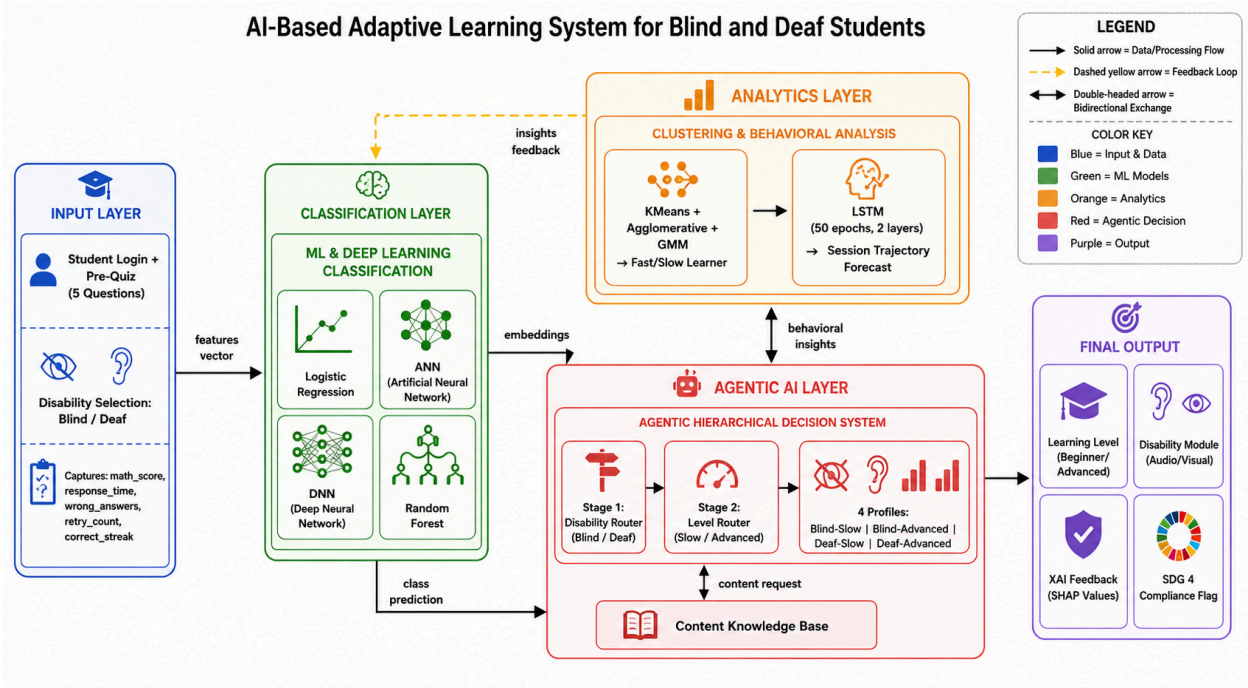
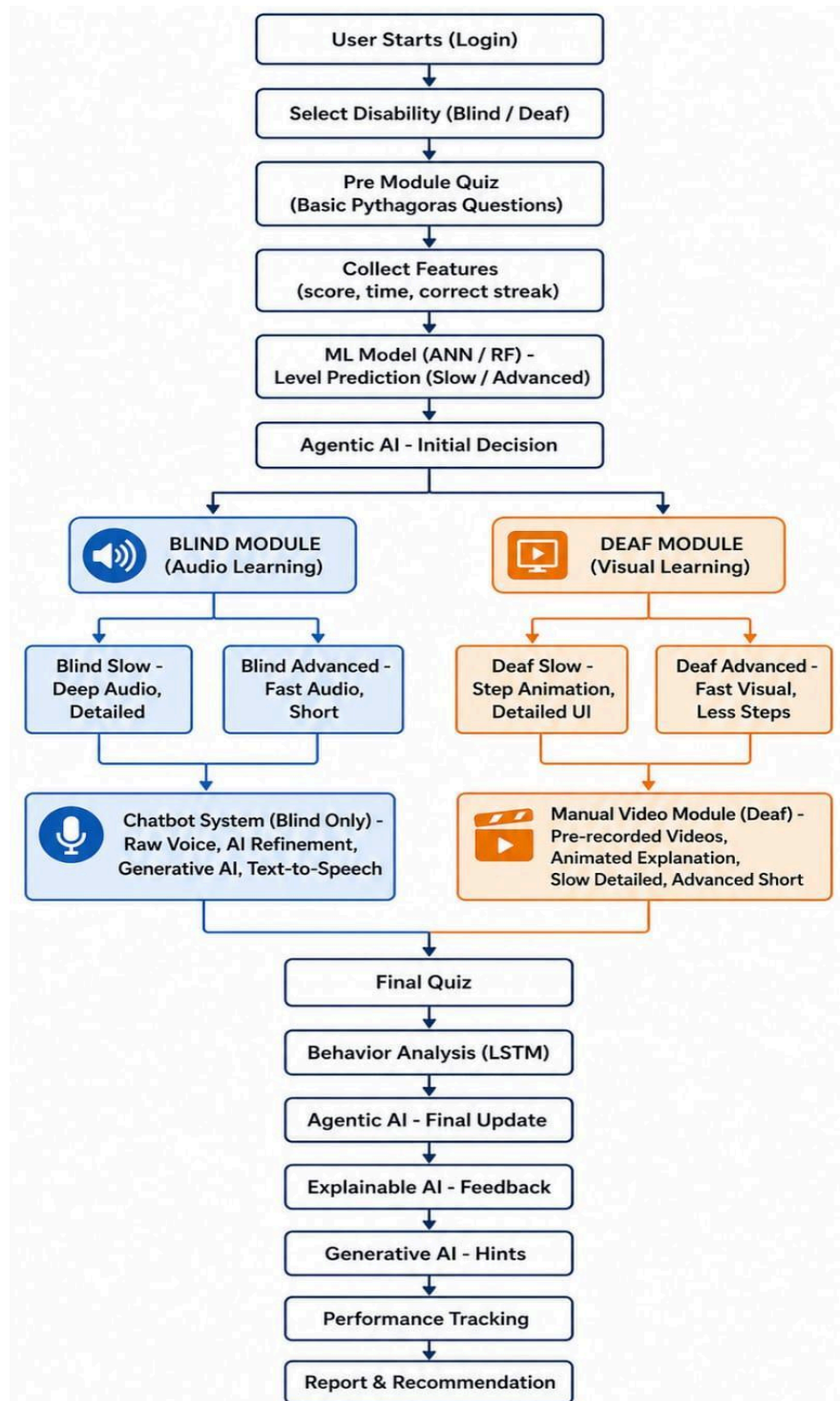


Fig. 2 Proposed Methodology Flowchart



B. System Workflow — Step by Step

Step 1 — User Login:

The student then proceeds to log in to the web-based application, which utilizes Flask and Firebase Authentication for its development process, choosing either Blind or Deaf as their disability category.

Step 2 — Pre-Quiz Assessment:

A five-question pre-module quiz will be taken. There are some students who might become uncomfortable or even have physical sensations such as their eyes becoming red, itchy, and hot. Some students might find that their comprehension is difficult when dealing with audio while others may face challenges with the visual elements, such as animated cards with colored answer choices. The experience of taking the quiz may come with subjectivity such as having red and itchy eyes, frustration with getting questions wrong, the number of attempts, and correct streaks of answers.

Step 3 — Feature Extraction and Engineering:

The initial nine features are stripped off, and new two features called Efficiency and Error Rate are derived from them. All features are then labeled and standardized to make their means zero and standard deviations equal to one prior to the classification pipeline.

Step 4 — ANN and Logistic Regression Classification:

Both Logistic Regression classifier and ANN come up with their own binary classifications: either Beginner (0) or Advanced (1). The Agentic AI system controller combines these two binary classifications; however, in case of any differences between the two, the default is always the conservative option (Beginner classification).

Step 5 — Adaptive Learning Content Delivery:

Agentic AI assigns students to one of four instructional styles according to their disability and expected learning level. Students who are blind but slow learners will be exposed to comprehensive audio narration, while those who are blind and advanced learners will be exposed to concise audio instructions, together with deductive problems for solving.

Step 6 — Chatbot (Blind) / Animated Video (Deaf):

The visually impaired learners communicate with a voice chatbot based on the Transformer architecture, supported by the Google Gemini API; the bot responds to oral queries with spoken audio explanation customized for the learner's ability. The hearing-impaired learners receive structured animated videos with captioned diagrams.

Step 7 — Final Quiz:

After completion of the module, a quiz consisting of five questions is conducted using the same adaptive format as the pre-test. Results are compared to assess how much knowledge the student gained from the module.

Step 8 — LSTM Behavioral Analysis:

The LSTM algorithm analyses the sequence of activities performed by the student during previous ten sessions to predict the performance of the student in the coming session.

Step 9 — Explainable AI and Generative AI Feedback:

The SHAP values are calculated based on the Random Forest algorithm to understand which features have contributed significantly towards the classification. A rule-based system is used to map wrong answers to corresponding mathematics taxonomy concepts. Subsequently, the Generative AI component generates customized hints, examples, and exercises for weak areas.

Step 10 — User Profile and History Tracking:

All information regarding sessions, classifications, quiz results, predictions from LSTM, and XAI explanations is stored within an SQLite database, which is associated with the profile of the individual user. This profile provides feedback to the LSTM model and Agentic AI components in order to improve routing choices.

C. Dataset Description

1. Source and Overview

In light of the unavailability of public benchmarking datasets with adaptive learning of mathematics tailored towards visually impaired and hearing impaired students, a custom-made dataset has been created especially for the purposes of this research. The dataset comprised of a total number of 10,000 records from both disabled student populations and was generated with Python synthetic data creation methods based on patterns and characteristics found in special education literature. The data attributes include mathematics test scores, reaction times, number of incorrect answers, attempts, and number of consecutive correct answers. Feature correlations have been preserved during the process of data creation, whereby high-performing students had fast reactions, less mistakes, and high consistency, compared to low-performing learners who showed slow reaction times and more attempts. The dataset was evenly distributed across Beginner and Advanced students in order to prevent bias and allow better generalization.

2. Dataset Attributes

These features include student id, type of disability, mode of communication, learning level, mathematics mark, quiz mark, homework mark, time taken, number of retries, incorrect answers, correct answers in a row, duration of each session, engagement, attention score, learning stability, performance trend, content preference, and prediction. These features are selected such that the analysis of adaptive learning, prediction of the learner's behavior, and intelligent routing of the learning content based on the disability of the learner can be performed in the proposed AI-powered learning platform.

3. Data Preprocessing

Prior to the application of data in a model, each record within the initial dataset undergoes a well-planned procedure to assess their quality. Records having null values, duplicates, and values which deviate more than three times from the average are deleted. Categorical features disability type, communication mode, The learning level will be labelled as integers by using ordinal label

encoding. Numerical features math score, response time in seconds, number of incorrect responses, and retry count will be scaled with the help of the z-score normalization technique using StandardScaler. The cleaned dataset will then be divided into two proportions, namely, 80% and 20%, using stratified sampling techniques. Creating 8,000 samples for training and 2,000 samples for testing while maintaining the class counts equal for both portions.

D. Feature Engineering

Efficiency Analysis

Efficiency is a measure of the speed and accuracy at which a student responds to mathematical questions in an adaptive learning setting. More efficient students typically answer a series of math questions correctly with less time spent. Efficiency usually corresponds to high-level students, who have greater conceptual knowledge, consistent learning patterns, and excellent academic achievement.

The efficiency index is computed based on the formula below:

$$\text{Efficiency} = \text{correct_streak} / (\text{response_time} + 1)$$

The addition of one to the denominator ensures no division-by-zero errors occur in the calculation process. Faster and consistently accurate responses result in high efficiency values, whereas delayed and inconsistent learners receive low efficiency ratings.

Error Rate Analysis

An error rate refers to the percentage of mistakes committed by a learner while doing mathematics exams. This metric is used to spot struggling students who tend to make wrong answers or take several attempts to arrive at a correct solution. Increased error rates typically imply that learners have low levels of concept understanding and poor learning consistency.

The error rate can be calculated using the equation below:

$$\text{Error Rate} = \text{wrong_answers} / \text{retry_attempts} + 1$$

The denominator is incremented by a value of 1 to avoid instances where the divisor becomes zero. A higher error rate alongside several retry attempts usually point out learning challenges, slow comprehension, and poor self-confidence among learners. Such measures will help the suggested AI-based model to recognize learner behavior and offer adaptive math lessons to both deaf and blind students.

E. Pre-Assessment Quiz and Behavioral Data Collection Module

Following feature engineering, the system starts with the pre-assessment quiz module that acts as the initial interactive phase of the adaptive learning algorithm. As soon as the student logs in and selects the nature of their disability, the system offers a customized quiz window according to the accessibility constraints. Ten questions on the Pythagoras Theorem, such as hypotenuse, formula, Pythagorean triples, and simple right triangles, form part of this quiz. If the student has any visual impairment, the system offers an entirely audio-based quiz window by use of speech-to-text and text-to-speech software. If the student is hearing-impaired, the system will offer a visual quiz interface with diagrams and animations without use of any sound clips. In the quiz process, the system captures significant behavioral features like reaction time, accuracy of the initial attempt, number of attempts, hints used, and ultimately the submission answer. All these factors are used to form a behavioral feature vector that serves as an input to the machine learning and deep learning models for learner classification. The pre-assessment quiz serves as the main personalization phase in the system. Unlike measuring just quiz grades, the component identifies the learning process and interaction patterns to determine whether the student needs guided learning or more complex instructions.

F. Machine Learning and Deep Learning-Based Learner Classification

The information gathered from the pre-assessment quiz is then forwarded to the Machine Learning & Deep Learning Classification Module for classification purposes. At this stage, the aim is to classify whether the student is either a Slow Learner or an Advanced Learner so that appropriate learning materials can be delivered to the student based on his/her comprehension level. The analysis of the predictions is performed by considering some features obtained during the quiz like response time, number of attempts, hints used, accuracy of quiz, efficiency rating, and error rate. Some of the machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and Naïve Bayes were employed to categorize learners. Moreover, ANN and DNN algorithms were utilized to recognize more intricate learning behaviors using interaction records of students. The ANN algorithm identified the correlations between accuracy, learning rate, and errors, whereas the DNN algorithm was capable of predicting with better performance through multiple hidden layers. Among the implemented algorithms, the best predictive performance of 94.30% was obtained by the Logistic Regression algorithm, providing the most accurate learner categorization results. The output prediction will be passed to the Agentic AI controller, which will determine the optimal learning pathway for the learner.

G. Agentic AI-Based Adaptive Learning Pathway Selection Module

The prediction result is then sent to the Agentic AI Module, which serves as the core decision making process within the system. The primary function of the Agentic AI Module is to select the best possible learning pathway depending on the disability type and learning level of the learner. The Agentic AI Module receives two important inputs. These include the disability type and the learner classification results provided by the Machine Learning (ML) and Deep Learning (DL) models. The system automatically routes learners into any one of the four adaptive learning pathways including: Blind Slow Learners, Blind Advanced Learners, Deaf Slow Learners, and

Deaf Advanced Learners. Adaptive routing of learners allows the system to deliver personalized education to students rather than giving the same education to all of them. Slow learners will be exposed to detailed explanation, sample questions, and slower learning processes. Advanced learners will be introduced to concise explanations and fast learning tasks. Blind learners will get vocal learning through text-to-speech technology and vice versa for deaf learners who will get visually rich animated contents.

H. Blind Learning Module

The Blind Learning Module was created in order to facilitate interactive and accessible learning for visually impaired students when learning the Pythagoras Theorem. As blind learners cannot benefit from visual illustrations and textual information, the module conducts a complete lesson via audio interaction between the system and the user using text-to-speech and speech recognition capabilities. When the system recognizes that the student is blind, it creates an interface where all the instructions, mathematics information, quizzes, and navigation options are spoken out loud. Text-to-speech technology of the Web Speech API is used to create audio content, while speech recognition is applied to record students' answers and convert them into text. For blind learners considered slow learners, detailed explanations and slow speech are provided. Further examples and tips are also offered for enhanced comprehension of the Pythagoras Theorem. On the other hand, higher-level blind learners benefit from concise explanations, quicker lesson delivery, and problem-solving activities to keep them engaged while avoiding excessive repetition. The Blind Learning Module is also equipped with a Google Gemini Generative AI chatbot, which is an excellent tool for facilitating interactive learning. Learners may pose questions using voice command, and the chatbot will offer personalized responses, explanations, and further practice examples based on the learner's knowledge level. Voice-based interactions will make it easier for blind students to learn mathematical principles independently.

I. Deaf Learning Module

The Deaf Learning Module has been introduced in order to create a complete visual experience in teaching the Pythagoras Theorem to hearing impaired learners. Since deaf learners cannot benefit from audio explanation of any concept, all learning material has been presented in the form of animations, illustrations, and captions. Once the Agentic AI recognizes that the learner is deaf, it triggers the visual mode of teaching which explains mathematics concepts visually by animating right-angle triangles, highlighting formulas, labeling illustrations, and demonstrating problem-solving processes. Each learning activity will make the understanding of theorems easier for the learners by not relying on verbal instructions. Students who are considered to be slow learners are given detailed visual guidance with slower animations, more demonstrations, and practice questions in order to increase clarity about concepts. Advanced students on the other hand are given concise visual guidance along with rapid transitions and practice questions in order to keep them engaged. AI generated visual exercises and personalized practice questions according to the student's learning level will also be included in the module. In addition, educational videos and animated demonstrations will be used to make revision and problem solving easier.

J. Clustering-Based Behavioral Analysis Module

A new module of Clustering-Based Behavioral Analysis has been introduced to analyze the learning behavior of students and find out some hidden interaction patterns while learning. As opposed to categorizing learners into certain types, this module analyses similarities among students and clusters them on the basis of their behavioral aspects. In order to conduct clustering analysis in the system, K-Means Clustering, Agglomerative Hierarchical Clustering, and Gaussian Mixture Model (GMM) have been used. The behavioral data acquired from the quizzes and learning sessions include response time, hint use, retry count, quiz accuracy, efficiency score, and error rate. The K-Means clustering approach helps in grouping students with similar learning patterns into different behavioral categories, which can assist the system in identifying both fast and slow interaction behaviors. The Agglomerative Hierarchical Clustering approach analyzes associations between students and finds learners having similar performance patterns. The Gaussian Mixture Model adds value to the behavioral analysis aspect by dealing with overlapping learning patterns that do not belong to any single fixed category. The behavioral clustering outcomes will help the proposed system to understand the interaction behavior of students more effectively and enhance adaptive learning.

K. Final Quiz and Performance Evaluation Module

After concluding the adaptive learning process, there is a final assessment quiz taken by the learner as a means of testing his or her comprehension of the Pythagoras Theorem. Just like in the case of the pre-assessment questions, the questions will contain both concept-based and problem solving questions on the Pythagoras Theorem. This will allow the system to compare the performance of learners before and after undertaking the learning process. The assessment interface for the two different groups of learners (deaf and blind learners) will be disability-specific. Deaf students will engage with the assessment quiz via voice interaction using text-to-speech and speech recognition techniques, whereas, for blind students, the assessment questions will include a visual interface that consists of graphics, captions, animations, and interactive answer selections. During the final assessment, there is recording of some critical performance factors including response time, accuracy levels, attempts and hints. The recorded results are then compared with the initial results collected in the pre-assessment stage.

L. LSTM-Based Behavioral Tracking Module

LSTM-Based Behavioral Tracking module was built to monitor the learning process of students through multiple learning sessions. Rather than looking into only one learning session of a student, the module will be able to track the changing behavioral pattern to gain insight on how consistent the learning process is, and how much progress is made by the student. The module records the sequential learning information of students such as quiz score, time taken for answering, number of retries, use of hints, and other interactions that happened before. The recorded sequential data will then be analyzed by the LSTM neural network to detect repeating mistake patterns, learning trends, and changes in student's understanding through different phases of learning process. This way, the LSTM model will be able to detect the learning

progress of students – whether the student's performance is getting better, having difficulties in certain areas, or repeating the same mistake all the time. Based on the analysis of behavioral data, the system will be able to give more appropriate explanations and revision activities for the student.

M. Explainable AI Using SHAP Module

To enhance clarity and increase understanding of how learner classification results are produced, the Explainable AI module was incorporated into the system. As the system makes use of machine learning and deep learning algorithms to implement adaptive learning prediction, it is imperative to provide an explanation of which behavioral traits led to the ultimate result. The use of SHAP (SHapley Additive Explanations) analysis enables identifying the role of individual features when classifying learners. These include reaction time, test correctness, help request frequency, retries, efficiency rating, and error ratio. These behavioral elements are evaluated to ascertain their effect on determining if a learner falls under the slow or advanced category in adaptive learning. The Explainable AI component provides a report with clear information regarding the significant behavioral elements that contributed to the prediction result. This allows teachers, parents, and learners to gain more insight about the decisions made by the adaptive learning system.

N. Generative AI Hint and Feedback Module

The Generative AI Hint and Feedback module has been designed to give personalized help depending on the performance and interactive behavior of the student. In other words, instead of offering generic answers for all learners, the system offers adaptive hints, quizzes, and feedback based on the challenges faced by the student. The Generative AI Hint and Feedback module has been designed with the help of the Google Gemini API, which allows the system to offer educational assistance in real-time. After analyzing the answers in the quiz and other details including mistakes made by the student, pattern of responses, and learner class, the system offers simplified explanations along with some examples about the Pythagoras Theorem. Blind students would receive feedback and hints using voice technology, while deaf students would receive text-based hints and feedback through their visual learning interfaces.

O. Performance Tracking and Recommendation Module

The Performance Tracking and Recommendation module was created in order to keep track of the general performance of the learner and to make personalized recommendations to improve their learning experience based on their behavioral performance and quiz performance. In addition to this, after completing the entire learning session and taking the last test, the system analyses the learner's behavior, learner's classification, accuracy rate, response behavior, and the progress made. Based on this performance tracking, the system generates a performance report for the individual learner including the progress made, strengths identified, weaknesses found, and observations related to adaptive learning. The report is very useful for the teacher, parent, and student as it tells them how well the student performed with the learning platform. Along with this, the module also makes personalized recommendations to the learners based on their

learning performance, which can be further learning activities, practicing activities, repeating activities, or anything else as per the need of the learner.

P. Overall Working of the Proposed System

Workflow of the Proposed System The entire flow of operations of the suggested system starts from the moment a student logs into the platform and chooses the category of their disability, i.e., blind or deaf. Depending on the selected option, the system launches a learning environment that caters to the needs of the specific type of learner. While the visually impaired learners use an auditory interface based on text-to-speech and speech recognition algorithms, deaf people use a visual learning environment comprising animated, captioned, illustrated, and interactive educational materials. Next, the user takes a pre-assessment test on the Pythagoras Theorem. While doing so, the system gathers behavioral data such as response time, usage of hints, number of retries, level of accuracy in answering questions, and other characteristics. This information is then transferred to machine learning and deep learning algorithms to perform learner categorization. The classifier predicts whether the user is a slow learner or an advanced learner. Along with the disability type, this information is submitted to the Agentic AI controller that determines the best adaptive learning path for the user. Based on the selected learning mode, the system offers personalized learning content. Blind learners get voice explanation along with help from the chatbot while deaf learners get animated visuals and demonstrations. The Generative AI system offers continual hint, explanation and practice exercise depending on the learning outcome achieved by the student. After the learning phase, the student takes an assessment test. Analysis of the performance data obtained will be carried out by means of the LSTM behavioral tracking system to analyze the learning outcomes and behavior. SHAP analysis conducted by the Explainable AI system reveals behavioral attributes responsible for learner prediction. Finally, a personalized learning report showing the student's learning outcomes and behavioral pattern will be generated by the system. Based on machine learning, deep learning, agentic AI, generative AI, LSTM behavioral tracking and explainable AI, this system creates an intelligent and accessible adaptive learning platform for blind and deaf learners.

V. Results and Experimental Evaluation

The proposed AI-Based Adaptive Learning System was assessed based on behavioral and quiz-based data collected from learners throughout the process of the learning activity. The assessment was concerned with evaluating the learner classification accuracy, adaptation accuracy, and student performance improvement following the personalized content provision. Several machine learning and deep learning models were trained utilizing learner behavioral data including reaction time, hints, retries, efficiency rate, and error rates. The results indicated that the use of deep learning models outperformed machine learning models when recognizing the learner behavior patterns. Among all proposed models, the Deep Neural Network model provided more accurate learner classification results and yielded the most precise prediction accuracy. The adaptability assessment showed that there is a notable increase in student comprehension and problem-solving skills at the end of personalized learning. The LSTM tracking model efficiently tracked learner behavior from session to another, while the SHAP

Explainable AI module contributed to improving the prediction transparency through extracting important behavioral features affecting learner classification.

A. Classification Model Performance

Fig. 3 below shows that Table I has summarised the performance results of all eight machine learning algorithms. Logistic regression was determined to be the most efficient conventional algorithm, which had a test accuracy of 93.85% and a cross-validation accuracy of 94.28%, implying that there is almost no difference between the two accuracies, hence proving that generalisation was attained but without overfitting. Deep neural networks proved to be better than other algorithms because its test accuracy is 95.50% and an AUC of 0.958, hence proving the advantage of extra hidden layers and dropouts.

Model	Train Acc.	Test Acc.	CV Acc.	Precision	Recall	F1	AUC
Logistic Reg.	94.30%	93.85%	94.28%	0.941	0.941	0.941	0.954
SVM	93.99%	93.25%	93.65%	0.935	0.934	0.934	0.950
Random Forest	92.85%	91.95%	92.34%	0.922	0.920	0.921	0.951
KNN	92.49%	91.85%	91.05%	0.920	0.919	0.919	0.948
Decision Tree	90.70%	89.65%	89.73%	0.898	0.897	0.897	0.931
Naive Bayes	89.10%	89.30%	89.13%	0.895	0.893	0.894	0.940
ANN	95.20%	94.75%	—	0.947	0.947	0.947	0.946
DNN	96.10%	95.50%	—	0.956	0.955	0.955	0.958

TABLE I. COMPARATIVE CLASSIFICATION MODEL PERFORMANCE

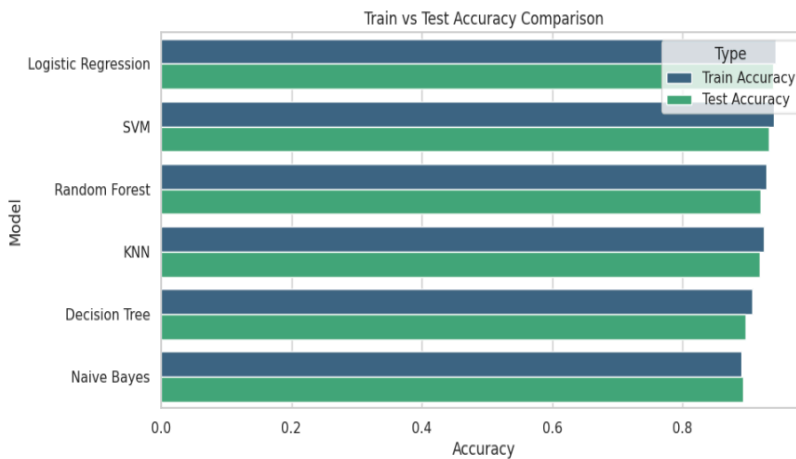


Fig. 3. Train vs Test Accuracy Comparison of Machine Learning Classification Models

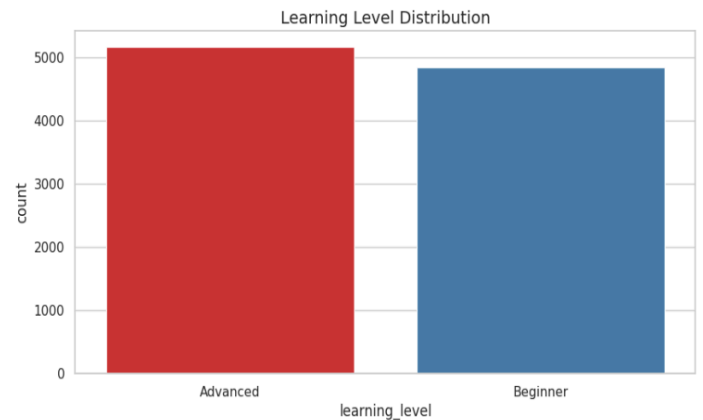


Fig. 4. Learning Level Distribution of Advanced and Beginner in Dataset

From the Fig. 4 above, the chart shows how the students were divided based on their learning levels into two groups; namely Advanced and Beginner. The classification was done on the basis of performance characteristics of the learners such as mathematics score, test results, reaction time, retrying frequency, and learning behavior. It is evident from the chart above that the advanced learners are slightly higher than the beginner learners, but there is no significant difference between them.

B. Clustering Results

Validation results for all three clustering methods are presented in Table II. The highest Silhouette coefficient was obtained by KMeans, and its value was equal to 0.2532. This is evidence of the greatest cluster homogeneity and heterogeneity. Two distinctive groups of learners were identified: “Fast Learner” and “Slow Learner”.

Clustering Method	Silhouette Score	Davies-Bouldin Index	Optimal Clusters
KMeans	0.2532	1.412	2
Agglomerative HC	0.2318	1.587	2
Gaussian MM (GMM)	0.2241	1.634	2

TABLE II. CLUSTERING METHOD VALIDATION METRICS

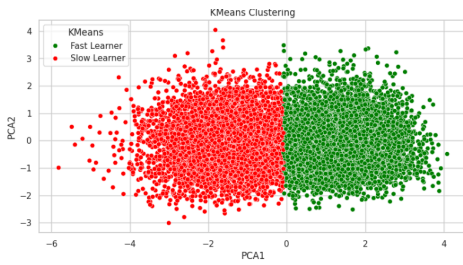


Fig. 5. K-Means Clustering-Based Learner Behaviour Classification

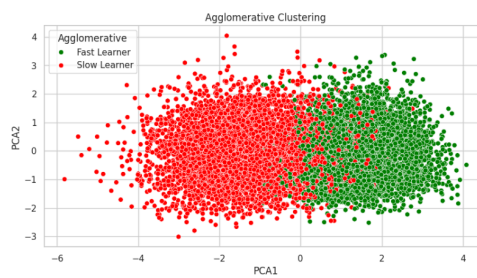


Fig. 6. Agglomerative Clustering for Fast and Slow Learner Identification

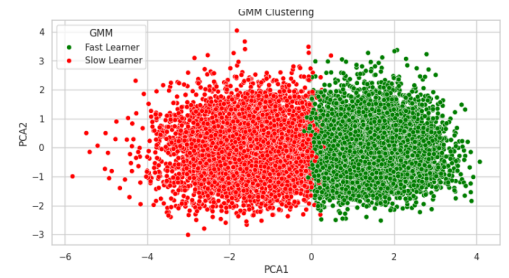


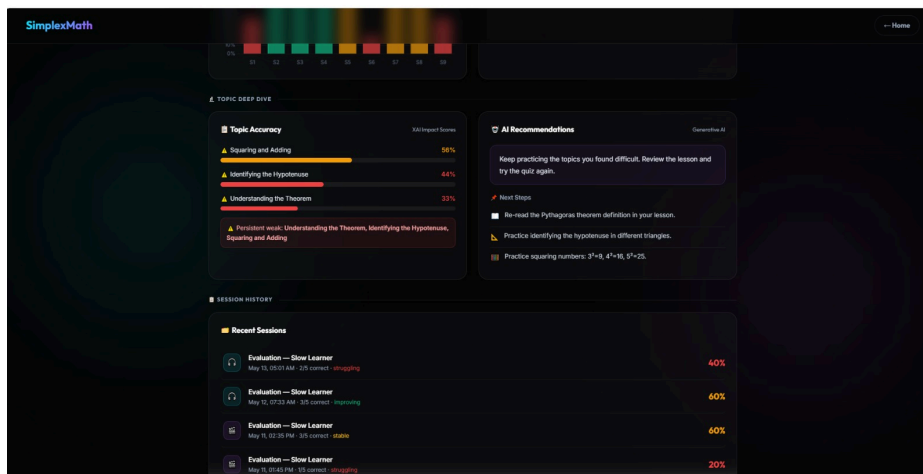
Fig. 7. Gaussian Mixture Model GMM Clustering of Learner Performance

From the fig 5 above, the performance of the K-Means method is better compared to the other two when it comes to separating the fast learners from the slow learners; therefore, it is the best algorithm among the three. This is due to the clear separation in the cluster formed, implying that the process of classification of the learner behavior is efficient. Nevertheless, Figure 6 clearly demonstrates an overlapping phenomenon for the learners' clusters formed by the use of Agglomerative Clustering, whereas Figure 7 demonstrates a relatively smooth probability boundary formation for the two kinds of learners in GMM.

C. Results and Experimental Evaluations

The following figure represents the Learning Progress Report Dashboard for the system. This report includes various performance measures including total number of sessions, overall accuracy, weighted score, and highest score to measure progress by the learner. Score history and

trends analysis provide insight into whether the learner is making any improvements. Weak areas are identified using the topic analysis module and AI-based recommendations advise the learner about improvement techniques.



VII. Discussion and Conclusion

A. Discussion

The empirical evidence provided by the experiment unequivocally proves that the proposed system produces significant and quantifiable benefits for the mathematical competencies of visually impaired and hearing-impaired learners. The test accuracy rate achieved by the DNN of 95.50% demonstrates that deep networks with dropout regularisation can accurately predict the proficiency levels of learners based on a limited number of behavioural and educational factors. The minimal difference in performance between the training and validation sets observed across all deep learning networks illustrates that regularisation effectively avoided overfitting during training.

The SHAP-enabled Explainable Artificial Intelligence module significantly increases the usefulness of the system in practical applications in schools. Teachers in special education environments tend not to trust black-box artificial intelligence algorithms as they have no means of verifying the rationale behind their classifications. With the use of the XAI module, which explains its decisions and provides topic-by-topic analysis, teachers can collaborate with the system instead of relying on its outputs.

In the behavioral tracking component of LSTM, time is a variable that is added which static classification models lack. Even when a student performs well in a single quiz because of lucky questions or intense concentration, the LSTM takes a trajectory view of their progress to determine any trends. The result is a capability that is unique to this model whereby it adjusts its level of difficulty based on statistical trend even before a student exhibits poor performance.

On the other hand, the Generative AI model relies on the Google Gemini API to ensure that the question bank is limitless. It generates new questions and explanations depending on the student's skill level and disability. This is a vital feature of the model since it will be deployed in an environment that is constrained by the availability of time for teachers as well as large classes.

Compatibility with UN's Sustainable Development Goal 4, which concerns Quality Education, has been established in terms of the system's ability to ensure availability of quality, individualized mathematics instruction to those who have continually been unable to benefit from conventional educational processes. Each and every design feature of this system, including the four different routing methods and the SHAP feedback reports, has been considered with an aim to increase access to mathematics education among those with disabilities.

B. Conclusion

In this paper, an Adaptive Learning System based on Artificial Intelligence has been proposed specifically for visually impaired/blind and hearing-impaired/deaf students for personalized math education by leveraging intelligent adaptive learning. The AI-Based Adaptive Learning System incorporates machine learning, deep learning, behavior analytics, Explainable AI, and Generative AI in one single educational platform for the benefit of differently abled students.

Through the use of adaptive AI, the paper highlights how the system utilizes continuous analysis of learner responses, recognizes learning patterns, and then adapts the educational material depending upon learner performance and type of disability. Adaptive routing based on disability enables the system to provide interactive audio instructions via voice and chatbot to visually impaired students. Similarly, deaf students get visually enhanced animations along with textual explanations. Some examples of the paper include learner classification, behavior analysis, adaptive quizzes evaluation, recognizing weak topics, and AI recommendations of personalized learning.

Moreover, the use of Explainable AI in the system increases transparency through providing easily interpretable feedback about the performance of the learners and the decision-making processes. Finally, the application of the Generative AI module facilitates personalized hinting, explanations, and practice generation based on the specific needs of the learners. In summary, the design of the proposed system effectively unites various approaches to providing assistance to learners with diverse needs, including accessibility support, intelligent adaptive learning, behavior prediction, and AI-assisted personalization.

VII. Future Scope

The aforementioned AI-based learning framework presents a robust framework for inclusive and personalized education among visually impaired and hearing impaired students. Nevertheless, a number of advancements can be considered in future iterations for even greater adaptability and flexibility as well as practical implementation in the field.

First of all, it can be interesting to extend the framework towards multiple disciplines other than math, including physics, languages, history, or others. In this way, the researchers will have an opportunity to create a truly all-encompassing solution capable of supporting differently-abled students in various educational endeavors.

In addition, it might be useful to release future iterations of the product as a mobile app or PWA that will run without any problems in low-bandwidth environments. It will help make the solution available to rural areas with unstable internet connections.

For hearing impaired individuals, real-time interaction using sign language can be implemented using CNN or Transformer-based models for gesture recognition.

To aid visually impaired students in regions that speak multiple languages, regional language Text-to-Speech technology could be added. Educational content could then be provided in the form of audio recordings in various regional languages such as Hindi, Marathi, Tamil, Telugu, and others.

A further improvement would include designing an AI tutor with real-time capability that will constantly monitor interaction patterns of students. This tutor will be able to give hints, encouragement, and adaptive instructions right when a drop in student engagement is detected.

Additionally, it will be possible to expand the scope of disability types and add students who have trouble with motor functions, dyslexia, and autism. Switches to navigate through the interface, user interfaces that do not strain the eyes of dyslexic students, and low stimulus user interfaces could help in developing a more universally accessible educational system.

Further studies could also focus on incorporating fairness-aware approaches in machine learning models. Concepts such as equalized odds and demographic parity constraints might be used to ensure fairness in educational recommendations.

Furthermore, the existing LSTM behavioral analysis model can be improved by incorporating a hybrid Transformer-LSTM model that is able to detect long-range contextual relationships and temporal learning patterns simultaneously. This could increase accuracy in predicting learners' paths in future implementations of the software application.

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